B.TECH. COMPUTER SCIENCE AND ENGINEERING 2022

THEORY OF COMPUTATION

Course Code: 23CSE403 **Credits: 03**
L: T: P: 2:1:0 **Contact Hours: 45**
Prerequisite(s): NIL

Course Objectives

The objective of the course is to

1. To Introduce concepts in automata theory and theory of computation
2. To Identify different formal language classes and their relationships
3. To Design grammars and recognizers for different formal languages
4. To Prove or disprove theorems in automata theory using its properties

Course Outcomes

At the end of the course, students will be able to

Course Outcomes	Description	Bloom's Taxonomy Level
CO1	Understand the characteristics of automata theory	Understanding (2)
CO2	Understand the FSA,DFA and NFA and its use	Understanding (2)
CO3	Apply and Relate the concept of the grammar with the concept of programming language	Applying (3)
CO4	Apply different Regular Expressions and its uses.	Applying (3)
CO5	Apply and solve the problems related to Finite Automata, RE, CFG, PDA and Turing Machine	Applying (3)
CO6	Present the theoretical preparation of programming languag s and compilers	Applying (3)

Course Contents

Module -1 **[09 hours]**

Introduction to Finite Automata: Introduction to finite automata, The central concepts of automata theory, Deterministic finite automata – definition, how a DFA processes strings, notations for DFA's Non-deterministic finite automata – definition, extended transition function, the language of an NFA